SOFTWARE QUALITY ASSURANCE TEST REPORT

RealWorld Demo Application | Manual Functional & Validation Testing

Tested application	https://demo.realworld.show
Test date	8 November 2026
Testing type	Manual functional, form-validation, authentication and profile-settings testing
Prepared by	QA Assessment Candidate
Test basis	Observed results supplied during the assessment testing session

1. Executive Summary

Manual testing was performed on the RealWorld demo hosted at **demo.realworld.show**. The original demo.realworld.io address was not accessible during the test session, so the alternative URL above was used. Seven distinct unexpected behaviours were observed and recorded. The findings mainly concern input validation, uniqueness checks for account identifiers, and password-strength enforcement.

Important: This report describes the observed behaviour of the tested demo instance. It does not establish the underlying source-code cause. Root-cause comments below are hypotheses to be confirmed by developers.

2. Scope and Test Summary

Area	Observed result
Account registration	Registration succeeded; duplicate email and duplicate username were accepted.
Authentication	Correct login succeeded. A later test confirmed an incorrect password was rejected.
Article management	Create, edit and delete operations succeeded. Whitespace-only title, description and body values persisted after refresh.
Comments	A whitespace-only comment was posted and remained visible after refresh.
Profile settings	Profile update succeeded; invalid email was rejected; duplicate email and username were accepted; password '123' was saved and could be used to log in.
Access control	Unauthenticated article editing was blocked; a second account did not see edit/delete controls for another user's article.

3. Defect Summary

BUG-001: Whitespace-only article fields are accepted | Severity: Medium

New Article form accepts a title, description and body containing only whitespace. The resulting article remained present after page refresh.

Steps to reproduce

1. Open the application and sign in.
2. Open New Article.
3. Enter spaces only in the title, description and body fields.
4. Publish the article and refresh the page.

Expected result: The application should trim input and reject fields that contain no meaningful characters.

Actual result: The article was published and remained visible after refresh, despite the fields containing only whitespace.

BUG-002: Whitespace-only comments are accepted | Severity: Medium

The comment form accepts a comment containing only whitespace, and the comment persists after refresh.

Steps to reproduce

1. Sign in and open an article.
2. Enter spaces only in the comment field.
3. Submit the comment and refresh the page.

Expected result: The application should reject a comment that is empty after trimming whitespace.

Actual result: The whitespace-only comment was posted and remained visible after refresh.

BUG-003: Registration accepts an email address already in use | Severity: High

A second account registration was successful when an email address already used by another account was entered with a different username.

Steps to reproduce

1. Register a first account.
2. Open a fresh Incognito window and choose Sign up.
3. Enter the same email address with a different username and password.
4. Submit the registration form.

Expected result: Registration should reject an email address that is already associated with an account and display a clear validation message.

Actual result: The website successfully created another account using the same email address.

BUG-004: Registration accepts a username already in use | Severity: High

A second registration succeeded when the username matched an existing account and a different email address was supplied.

Steps to reproduce

1. Register a first account.
2. Open a fresh Incognito window and choose Sign up.
3. Enter the same username with a different email address and password.
4. Submit the registration form.

Expected result: Registration should reject a username that is already assigned to another account.

Actual result: Registration was successful with the duplicate username.

BUG-005: Profile settings accept an email address used by another account | Severity: High

The profile settings form saved an email address that was already associated with another test account.

Steps to reproduce

1. Create two test accounts with different email addresses.

2. Sign in to the first account and open Settings.

3. Change its email to the second account's email address.

4. Select Update Settings.

Expected result: The application should prevent an account from changing its email to an address already used by another account.

Actual result: The website saved the email address already used by another account.

BUG-006: Profile settings accept a username used by another account | Severity: High

The profile settings form saved a username already assigned to another test account.

Steps to reproduce

1. Create two test accounts with different usernames.

2. Sign in to the first account and open Settings.

3. Change its username to the second account's username.

4. Select Update Settings.

Expected result: The application should prevent two accounts from having the same username.

Actual result: The website successfully saved the duplicate username.

BUG-007: Profile settings accept a very weak password | Severity: High

The settings form accepted the password '123'. After logging out, the tester successfully logged in again using that password.

Steps to reproduce

1. Sign in to a test account and open Settings.

2. Enter '123' in the New Password field.

3. Update Settings and log out.

4. Open a fresh Incognito window and sign in using the test account and password '123'.

Expected result: The application should enforce a reasonable minimum password-strength policy and reject a very weak password.

Actual result: The password '123' was saved, and login using that password succeeded.

4. Root Cause Analysis (Preliminary)

Selected issue: BUG-003 – duplicate email accepted during registration.

The observed behaviour suggests that duplicate-email validation may be missing or may not be enforced correctly during registration. The user interface may submit the registration request without checking whether the email is already registered. The backend may also be missing a uniqueness check for the email field. Another possibility is that the database does not enforce a unique constraint on the email column. If validation exists only in the frontend, it can be bypassed by sending a request directly to the backend. The same general weakness may explain why duplicate emails were accepted in profile settings, although this has not been confirmed from source code. Developers should inspect the registration and profile-update API handlers, validation logic, and database constraints. A server-side uniqueness check and a database unique constraint should be applied, with a clear error returned to the user. This is a preliminary hypothesis based on black-box testing, not a verified code-level root cause.

5. Passed Checks / Correct Behaviour Observed

• Empty registration form was blocked.

• Invalid email during registration was blocked.

• Login with an incorrect password was rejected (the earlier contrary observation was superseded by this later retest).

• Invalid email in profile settings was rejected.

• Profile settings update saved successfully.

• Logout returned the user to a public page.

• Unauthenticated article editor access was blocked.

• Unauthenticated article editing was blocked.

• A second account did not see Edit/Delete controls for another user's article.

• Create, edit and delete article operations completed successfully.

• Empty article submission and empty comment submission were blocked.

6. Recommendations

1. Trim user input on the server before validating article and comment fields.
2. Enforce unique email and username values on the backend and in the database.
3. Apply a documented password policy, including a minimum length and strength checks.
4. Add automated regression tests for registration, profile updates, article creation and comments.
5. Return clear validation messages without exposing sensitive account information.
6. Retest all fixes in a clean browser session and capture evidence screenshots.

7. Limitations and Evidence

This report is based on manual test outcomes reported during the assessment session. Screenshots, browser version, operating system, test account identifiers and API/network logs were not supplied for inclusion. Add redacted screenshots for each defect before final submission. Do not include passwords, authentication tokens, or private personal information in screenshots or the public repository. The tested URL was demo.realworld.show; results should not be attributed to demo.realworld.io.

End of Report